HO CHI MINH CITY DEPARTMENT OF EDUCATION & TRAINING

LAWRENCE S. TING SCHOOL

RESEARCH REPORT

OIL SPILL TREATMENT CAPABILITY OF CHICKEN FEATHER-DERIVED ADSORBENTS

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Field:

Materials Science

Research Location:

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DECLARATION

We hereby declare that this research report on the topic "Investigation and Evaluation of the Capability of Treating Oil Spills Using Adsorbent Materials Made from Chicken Feathers" is the research work of our group, carried out under the guidance of Ms. Lê Nguyễn Thảo Trang over the past period. All data used for analysis in the report and the research results were gathered and analyzed by ourselves objectively and honestly, are of clear origin, and have not been published in any form. We take full responsibility if there is any dishonesty in the information used in this research work.

Ho Chi Minh City, October 2024

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CHAPTER 1: INTRODUCTION

1.1. Research Purpose

According to the 2023 oil tanker spill statistics of the International Tanker Owners Pollution Federation, the total amount of oil released into the environment in 2023 exceeded 2000 tons [1]. Vietnam currently ranks third worldwide in the number of oil spill incidents. From 1992 to 2019, 190 oil spill incidents occurred in Vietnam (37 offshore cases, or 19%; 88 coastal cases, or 47%; and 65 cases, or 34%) [2]. Oil spills can reduce the regenerative capacity of ecosystems such as mangrove forests, seagrass beds, sandy tidal flats, lagoons, and coral reefs. Oil on the sea surface reduces oxygen exchange between the air and the water, unbalancing the oxygen content of the ecosystem. In addition, heavy metals, sulfur, and other components settle and accumulate on the seabed, causing pollution, and a large concentration of spilled oil can cause mass fish deaths from oxygen deficiency or poisoning. In the long term, this can create a shortage of individuals in the food chain of ecosystems and cause an imbalance. The tourism and fishing industries are also severely affected because spilled oil spoils the scenery and damages machinery, hindering fishing [3]. In fact, in the cargo ship collision that caused an oil spill on the Lòng Tàu River, Cần Giờ District (Ho Chi Minh City) in April 2025, more than 4 tons of farmed fish were recorded dead as a direct result of the spilled oil, and the total area of land and forest affected by the spill was 30.9654 ha [4].

Chicken feathers are a by-product of the poultry farming and processing industry; an estimated 1000 tons of chicken feathers are discarded each week in the UK and 10000 tons each day worldwide [5]. Chicken feathers are composed largely of keratin, a hard, fibrous protein. Keratin is insoluble in water and not degraded by common proteases, which gives chicken feathers important properties such as water resistance, resistance to natural decomposition, and high mechanical strength [6]. In addition, chicken feathers are composed of keratin – a protein with high mechanical strength and resistance to biodegradation – and have a characteristic hollow honeycomb structure. This structure creates a large surface area and many voids, enabling chicken feathers to adsorb and retain considerable amounts of liquids, including petroleum and other organic compounds [7]. Therefore, using chicken feathers as a raw material to fabricate materials for treating oil spills is considered a highly effective solution.

From the above, researching and applying chicken feathers to make oil spill treatment materials has scientific significance and contributes practically to environmental protection and sustainable development. That is why we chose the topic “Investigation of the Capability of Treating Oil Spills Using Adsorbent Materials Derived from Chicken Feathers”.

1.2. Research Tasks

- Task 1: Fabricate adsorbent materials made from chicken feathers
- Task 2: Investigate and evaluate the adsorption capacity of chicken feather adsorbents
- Task 3: Enhance the stability of the material in aqueous environments by combining it with a biological binder

1.3. Research Subject and Scope

1.3.1. Research Subject

Adsorbent materials made from chicken feathers

1.3.2. Research Scope

Spatial scope: Ho Chi Minh City, Vietnam
Time scope: 10/2024 – 09/2025
Content scope: Investigation and evaluation of the adsorption capability of adsorbent materials made from chicken feathers in Vietnam

1.4. Novelty of the Study

Many studies have made adsorbent materials from animal hair and feathers (human hair, sheep wool, chicken feathers, duck feathers…) and natural sources such as sugarcane bagasse and water hyacinth,… Among them, chicken feathers show high oil adsorption capacity, which varies with chicken breed and treatment process: the Baladi breed (Egypt) reached up to 13770 mg/g (crude oil) and 11990 mg/g (diesel) [8]. Meanwhile, local chickens in the UK reached only 6059 mg/g and 4097 mg/g [9]. Chicken breeds in different regions thus differ in adsorption capacity, depending on each breed's characteristics and the production process of the adsorbent material. Currently in Vietnam, no study has combined poultry feathers with a biological binder to ensure the material's durability and stability in treating oil spills. In Vietnam, no study has combined poultry feathers with a biological binder to increase durability and stability, laying the groundwork for developing sustainable oil spill treatment materials from livestock by-products.

1.5. Research Questions

- Can adsorbent materials made from chicken feathers adsorb oil?
- Are adsorbent materials made from chicken feathers durable in oil?

1.6. Scientific Hypothesis

Chicken feathers contain water-resistant keratin and have a porous hollow structure, favoring oil adsorption. When combined with kapok powder, the material has good recovery capability and the product has high mechanical strength, is easily retrieved, and is environmentally friendly. Its adsorption capacity is assumed to depend on soaking time and mass: the longer the contact time and the greater the mass, the more oil is adsorbed. Hence, a material combining chicken feathers and kapok powder is hypothesized to be an effective, sustainable oil spill adsorbent that can be retrieved while making use of livestock by-products.

CHAPTER 2: THEORETICAL BASIS

2.1. Background of the Research Problem

Environmental pollution from industrial and agricultural activities, especially heavy metals and dyes, has become a global concern [10]. Adsorption is an effective method for wastewater treatment [10]. However, traditional adsorbents are not economically efficient when applied on a large scale [10]. Therefore, researchers have turned to finding adsorbents that are readily available and non-toxic [10]. Chicken feathers, a by-product discarded in large quantities by the poultry industry (about 8.5 billion tons per year worldwide*)*, not only help reduce waste but also provide an effective and environmentally friendly oil adsorbent [8], [10], [11], [12].

According to previous studies, Baladi chicken feathers showed an oil adsorption efficiency of 98% with 20 ml of oil per 1 g of feathers and 85.3% with 80 ml of oil per 1 g of feathers [8]. When chemically treated, this efficiency rose to a maximum of 99% (20 ml of oil) and a minimum of 95.5% (80 ml of oil) per 1 g of feathers [8]. The maximum adsorption capacity recorded was no more than 120 ml (equivalent to 105 g) of oil per 1 g of feathers [8]. Fluffy chicken feathers had the highest oil adsorption capacity, about 18.9 ± 0.5 g oil/g feathers in a 15-minute test and 20.5 ± 1.9 g/g in a 24-hour test [11]. For different oils, chicken feathers adsorb about 13.10 ± 0.67 g/g of crude oil, 11.15±0.84 g/g of diesel oil, and 9.70 ± 0.91 g/g of kerosene [12] Furthermore, more than 85% of the adsorbed oil can be recovered from chicken feathers, more than with synthetic adsorbents [12], [13]. This shows that chicken feathers are an oil adsorbent for treating oil pollution on land and in aquatic environments [12].

2.2. Adsorption Capacity of Chicken Feathers

2.2.1. Structure of Chicken Feathers

Chicken feathers contain about 91% keratin, which provides reactive sites that can naturally bind metals, organic compounds, and gases [10]. Chicken feathers are predominantly hydrophobic, with a 60:40 ratio in favor of hydrophobic amino acids, and cysteine disulfide bonds help them attract oil and repel water [8], [11], [13]. Morphologically, chicken feathers have a unique structure of rachis, barbs, and barbules [10], [11]. In particular, the barbs have a hollow honeycomb-like cross-section, creating internal cavities that hold liquid and give chicken feathers a low density (about 0.8 g/cm³), acting as a superabsorbent material for oil [10], [11], [13]. In addition, chicken feathers are classified as mesoporous and microporous materials, which increases surface area and adsorption capacity [12], [13].

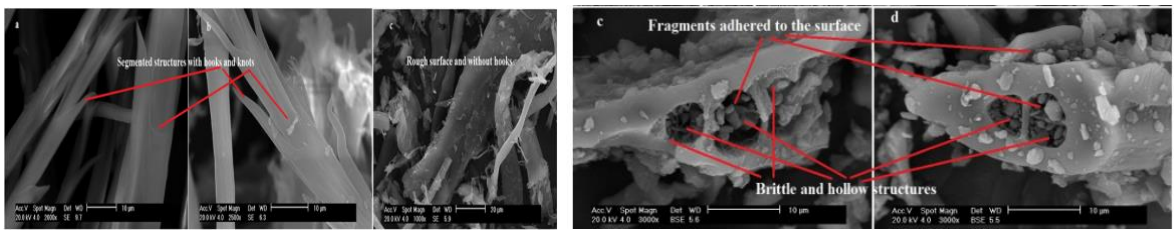


Figure 2.2.1. Hollow structure of chicken feathers [14]

2.2.2. Oil Adsorption Mechanism of Chicken Feathers

Studies show that the oil adsorption process onto chicken feathers best fits the Langmuir isotherm model and the pseudo-second-order kinetic model [12], [15]. This demonstrates that the adsorption process is monolayer, takes place on sites with similar adsorption energy, and is chemical in nature [10], [12], [15].

Chicken feathers consist mainly of keratin (about 91%), along with 8% water and 1% lipids [8], [10]. Keratin is a special fibrous protein in which hydrophobic amino acids predominate in the polypeptide chain, and the presence of strong disulfide bonds from cysteine gives the keratin in chicken feathers its hydrophobic properties [10], [11], [12], [16]. Keratin also improves the tensile strength, structural durability, water insolubility, and stability over a wide pH range of chicken feathers [10]. This structural durability is further enhanced by hydrogen bonds between segments of amino acid chains [10].

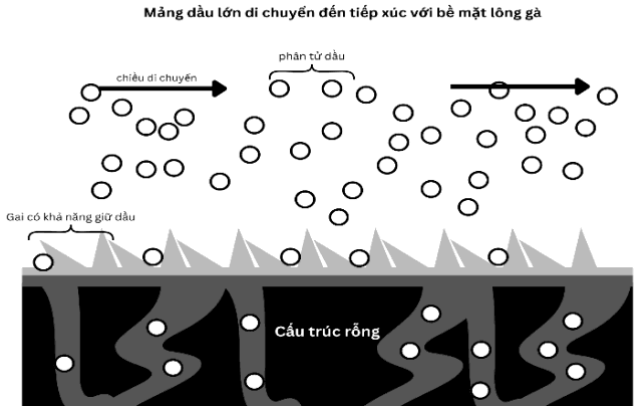


Figure 2.2.2. Adsorption model on the surface of chicken feathers [10]

2.2.3. Binding Material for Chicken Feathers – Kapok Powder

Kapok leaves are dried and finely ground into powder, called kapok powder. Kapok leaves have been studied and found to contain the nutrients needed to feed aquatic species. Industrially raised fish need a balanced diet to grow optimally [17]. All the nutrients that fish need are present in kapok leaves: protein, fat, minerals, vitamins, and fiber [17]. Besides these nutrients, kapok leaves also contain other phytochemicals with medicinal properties that act against heart disease, cancer, or diabetes [18]. Moreover, mixing kapok powder with water yields a highly adhesive mixture. From the above characteristics, kapok leaves have been shown to contain many necessary nutrients and to be safe for the organisms that consume them, and can be used as a binder for chicken feather fragments, making them suitable for fabricating effective oil spill adsorbent materials.

CHAPTER 3: RESEARCH METHODOLOGY

3.1. Materials and Equipment

Chemicals	Equipment
Chicken feathers	Drying oven
Kapok powder	Analytical balance
5% $NaHCO_3$ solution	Overhead stirrer
Javel solution	
DO 0.001S-V diesel oil	
Distilled water	

Table 3.1. Materials and equipment

3.2. Research Procedure and Methods

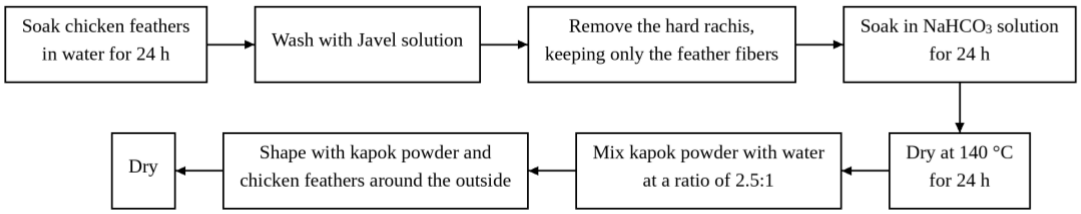


Figure 3.2. Overview of the process for making adsorbent materials from chicken feathers

3.3. Evaluation Criteria and Experimental Methods

3.3.1. Investigation of Oil Adsorption Capacity

3.3.1.1. Effect of Mass

To investigate the oil adsorption capacity, chicken feather samples (0.2, 0.4, 0.6, 0.8, and 1.0 g) and kapok powder samples (0.2, 0.4, 0.6, 0.8, 1.0, 1.2, and 1.4 g) were prepared and soaked in oil for 1 hour. The samples were then removed, left to drain for 1 hour, and reweighed. The experiment was carried out at room temperature (25 ± 2 °C), repeated 3 times for each condition, and the mean value was used for analysis.

Sample	G0.2	G0.4	G0.6	G0.8	G1.0
Mass (grams)	0.2	0.4	0.6	0.8	1.0

Table 3.3.1.1.1. Mass composition of chicken feathers in each sample

Sample	*BG0.2*	*BG0.4*	*BG0.6*	*BG0.8*	*BG1.0*	*BG1.2*	*BG1.4*
Mass (grams)	*0.2*	*0.4*	*0.6*	*0.8*	*1.0*	*1.2*	*1.4*

Table 3.3.1.1.2. Mass composition of kapok powder in each sample

3.3.1.2. Effect of Time

To investigate oil adsorption capacity over time, chicken feather samples (0.6 g) and kapok powder samples (1.2 g) were soaked in 30 mL of oil. The mass was measured before and after soaking at 10, 20, 30, 40, 50, and 60 minutes. After each time point, the mixture was filtered through filter paper for 1 hour, and the chicken feathers and kapok powder were reweighed. The experiment was carried out at room temperature (25 ± 2 °C), repeated 3 times for each condition, and the mean value was used for analysis.

3.3.2. Investigation of Durability in an Oil Environment

Samples of chicken feathers (0.6 g) and kapok powder (1.2 g) were prepared and then soaked in oil at room temperature (25 ± 2 °C). Changes in the morphology and condition of the samples after certain time intervals were recorded by visual observation. The experiment time was recorded at time points every 24 hours throughout 7 days.

CHAPTER 4: RESEARCH RESULTS

4.1. Oil Adsorption Capacity

4.1.1. Effect of Mass

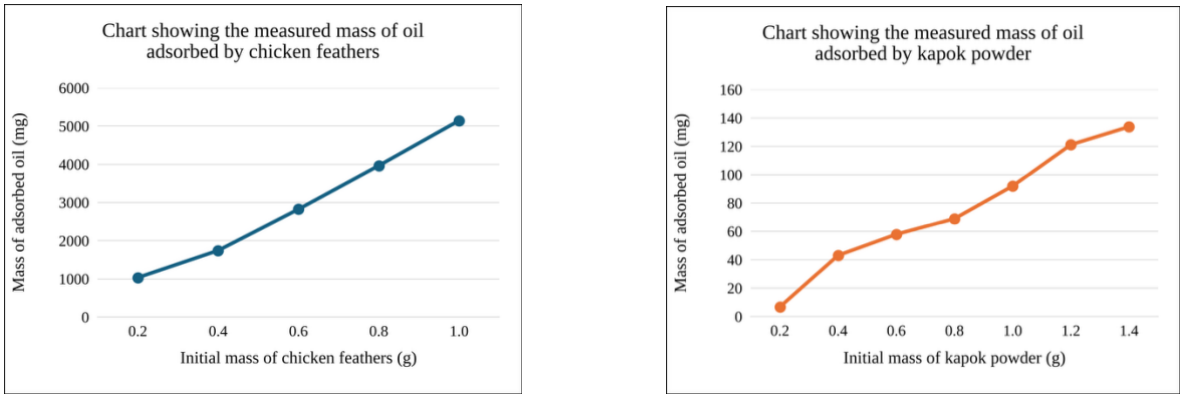


Figure 4.1.1. Oil adsorption results of chicken feathers and kapok powder by mass

From the charts, the oil adsorption capacity of both materials increases with the mass of material used, showing a clear directly proportional relationship. Chicken feathers show a very strong oil adsorption tendency, with the amount of oil retained increasing steadily from more than 1000 mg at 0.2 g to over 5000 mg at 1.0 g, indicating an oil adsorption capacity of up to 500%. This demonstrates that chicken feathers are a highly effective adsorbent, owing to their characteristic keratin structure, which is rich in pores and hydrophobic, helping to retain a large amount of oil. Thus, chicken feathers not only tend to adsorb oil well but also have great potential for sustainable, environmentally friendly oil spill treatment.

4.1.2. Effect of Time

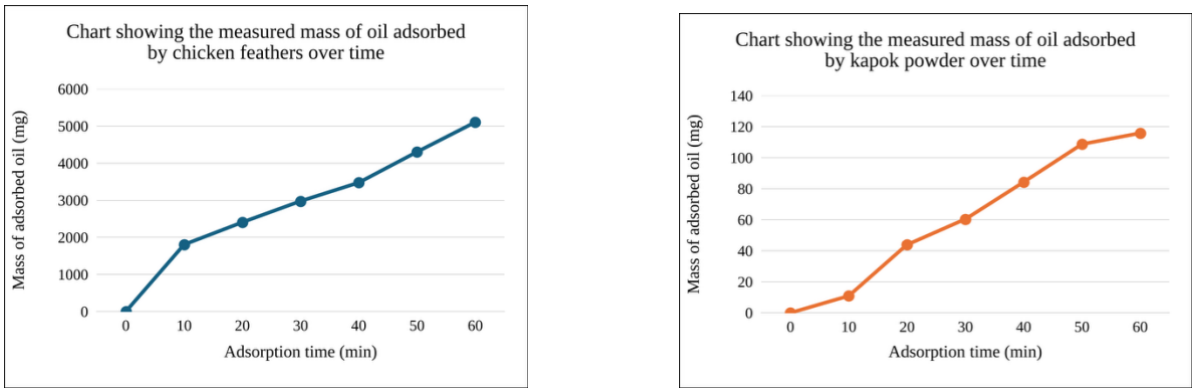


Figure 4.1.2. Oil adsorption results of chicken feathers and kapok powder over time

The charts show that the mass of oil adsorbed by chicken feathers increases gradually over time. In the initial stage (0 – 10 minutes), the adsorption rate is fast, reflecting very effective contact and occupation of the material surface by oil molecules; from 20 – 40 minutes, adsorption continues to increase but slows as the available adsorption sites are gradually filled; by 50 – 60 minutes, the amount of oil adsorbed keeps rising and approaches saturation, indicating that the material has nearly reached its maximum adsorption capacity. Thus, chicken feathers can be

concluded to have good oil adsorption capacity, with adsorption proceeding continuously over time and stable oil retention, confirming their potential as a natural, environmentally friendly, and effective adsorbent for treating oil spills.

4.2. Investigation of Durability in an Oil Environment

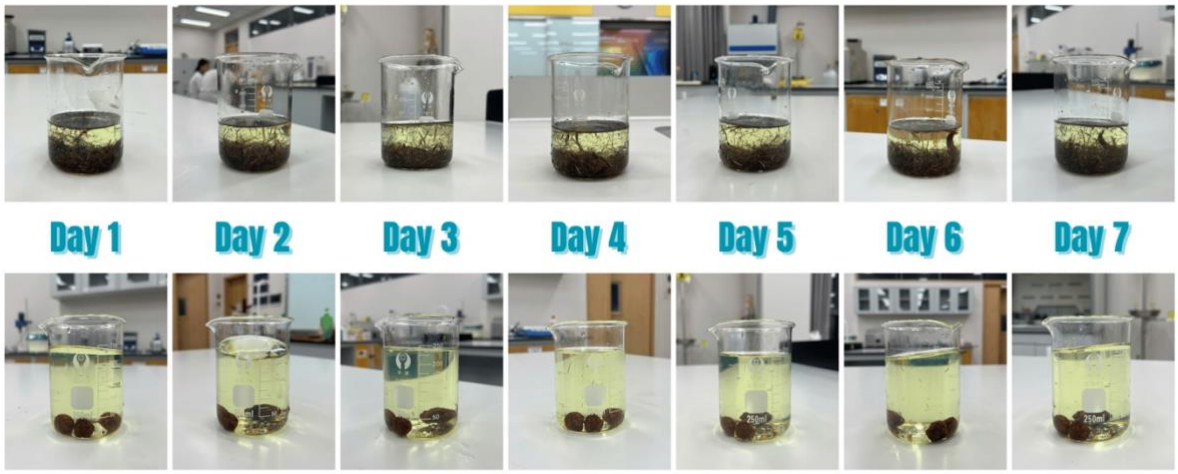


Figure 4.2. Durability test results of chicken feathers and kapok powder in oil

After 7 days of soaking in oil, the chicken feather samples showed no decay, breakage, or significant structural deformation. Observation showed only slight changes in gloss and softness, while mechanical and physical properties remained stable. Similarly, the kapok powder samples showed no signs of decomposition, cracking, or marked loss of mechanical properties. The results confirm that both chicken feathers and kapok powder maintain durable mechanical properties during prolonged contact with oil, demonstrating their potential for real-world applications.

4.3. Investigation of the Combined Chicken Feather and Kapok Powder Adsorbent

4.3.1. Investigation of Oil Adsorption Capacity over Time

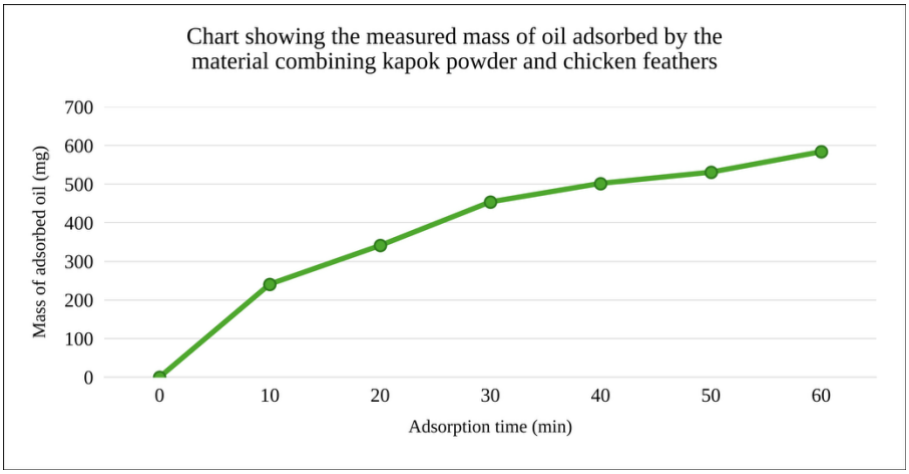


Figure 4.3.1. Oil adsorption results of combined chicken feather and kapok powder over time

The chart shows that the mass of oil adsorbed by the combined kapok powder and chicken feather material increases gradually over time. In the first 30 minutes, adsorption increases rapidly, showing that the material works effectively at the initial stage. From 30 minutes onward, the adsorption rate slows and tends toward saturation at around 60 minutes. Thus, the material has good oil adsorption capacity, especially at the beginning, and the optimal adsorption time for high efficiency is 30 to 40 minutes.

4.3.2. Investigation of Durability

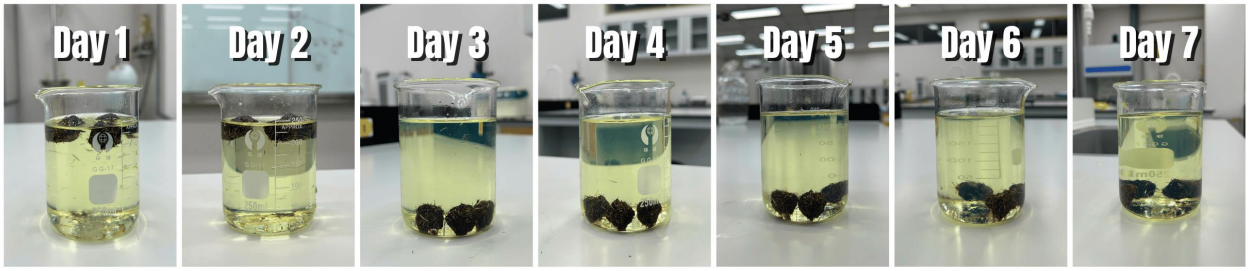


Figure 4.3.2. Durability test results of the adsorbent material in oil

The results of the durability investigation of the adsorbent material combining chicken feathers and kapok powder in oil show that the material remained stable and effective at absorbing oil throughout the 7-day test. Throughout the observation period, the oil showed no clear change in physicochemical properties, color, or transparency. The material also showed no signs of decomposition or structural degradation. These results demonstrate that the combined chicken feather and kapok powder material is highly durable and can remain effective in oil for a long time without being significantly affected.

CHAPTER 5: CONCLUSION AND DISCUSSION

5.1. Conclusion

Through the experiments investigating the oil adsorption capacity and durability of this material, based on the promising results, we can conclude that:

Chicken feathers and the kapok powder mixture adsorb oil effectively, and oil adsorption depends more on the mass of chicken feathers than of kapok powder.

Kapok powder and the material combining kapok powder with chicken feathers are durable in an oil environment and do not disintegrate over a period of 7 days.

5.2. Discussion

After carrying out the research, we make the following recommendations:

- Measure additional samples with different kapok powder to chicken feather ratios to diversify the data for analysis and propose more practical solutions.
- Evaluate the reusability of the material by measuring the amount of oil retained after using the centrifugation method.
- Investigate the oil adsorption capacity using the BET measurement method
- Increase samples and time points to verify the oil adsorption capacity of the material
- Investigate the oil adsorption efficiency of the material in a simulated seawater environment
- Test the application and recovery of the adsorbent material in a marine environment over a reasonable period of time
- Develop a method to quickly produce round, uniform kapok powder samples of equivalent mass.

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